

LITTLE STREAM DETECTOR

LSD 3.0

Detailed Technical Documentation

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Release	3.0 Final Release
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External multimedia tools	None
Temporary media files	None

Scope. This document describes the native container, codec, canonical-sample, analysis, validation, reporting, threading, safety, and dual-result comparison architecture of LSD 3.0. Version 3.0 preserves the canonical analysis core introduced in 2.0 and adds a reference-comparison presentation layer built from immutable result snapshots.

Release principle

The 3.0 feature set does not create a second parser path. A previously completed result can be promoted to Reference B, while a later analysis becomes Current A. Existing bitrate and quantizer results are then rendered together with fixed role colors and independent summaries.

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1. Purpose and System Scope

Little Stream Detector 3.0 is a native Windows PowerShell application with an embedded C# analysis engine. It inspects media containers, builds a container-neutral sample index, performs codec-aware bitstream analysis, and presents both human-readable and structured results through a responsive graphical interface.

Version 3.0 adds a one-reference comparison workflow. The first completed analysis can be promoted from Current A to Reference B. A later file is analyzed through the same pipeline and becomes a new Current A. Both completed result sets can then be compared without rerunning or duplicating container and codec logic.

1.1 Primary objectives

- Provide deterministic native analysis without FFmpeg, MedialInfo, or other external multimedia processes.
- Keep container-specific addressing separate from codec-specific interpretation.
- Validate every canonical sample against file boundaries before reading payload data.
- Report unavailable information as N/A rather than guessing.
- Support asynchronous preparation and analysis without blocking the GUI.
- Retain Windows PowerShell 5.1 and in-memory EXE hosting compatibility.
- Allow visual comparison of exactly one Current A and one Reference B using completed native results.

1.2 Supported families

Area	Supported families
Containers	Matroska, WebM, AVI 1.0, OpenDML AVI 2.0, MP4, M4V, MOV
Video	H.264/AVC, H.265/HEVC, AV1, MPEG-4 Part 2/XviD
Audio	AAC-LC, HE-AAC, HE-AACv2, MP3, PCM integer/float, AC-3, E-AC-3
Outputs	Summary A, optional Summary B, Streams, JSON, Log, bitrate profile, quantizer distribution

2. Architectural Overview

LSD remains a modular monolith. PowerShell owns application state, presentation, orchestration, and background-job lifecycle. Embedded C# types own binary parsing, canonical models, bitstream traversal, validation, and numerical accumulation.

```
User / GUI | Metadata and job coordinator | +-----+-----+-----+ | | | Matroska MP4 AVI/OpenDML
+-----+-----+ | Canonical media model | Video + Audio analyzers | Completed result snapshot | |
Current A Reference B +----- comparison views
```

2.1 Logical layers

Layer	Responsibility	Representative components
Presentation	Window, controls, reports, graphs, drag-and-drop, active-tab copy	PowerShell WinForms
Coordination	Metadata preparation, state transitions, cancellation, progress	Start-Meta, Complete-NativeMetadata, Start-Scan
Container parsing	Detect structure, read tracks, resolve sample locations	NativeMediaProbe, NativeMp4Probe, NativeAviProbe
Canonical adaptation	Normalize tracks and samples	MatroskaToLsdAdapter, Mp4ToLsdAdapter, AviToLsdAdapter
Codec analysis	Parse codec configuration and frame-level syntax	AVC, HEVC, AV1, MPEG-4 Visual, AAC, MPEG Audio, Dolby, PCM
Comparison	Retain Reference B and render A/B result views	ReferenceResult, ReferenceOnlyView, comparison renderers

2.2 Canonical pipeline continuity

The 3.0 comparison layer consumes completed output from the same canonical pipeline used by 2.0. It does not alter sample addressing, codec parsing, validation, or completeness rules. Internal adapter names retain the /2.0 suffix because they identify the stable canonical adapter contract, not the application release number.

3. Runtime and Processing Lifecycle

3.1 Startup

- Compile embedded C# through Add-Type.
- Run native self-tests for selected bit-reader and arithmetic behavior.
- Hide the console window for GUI operation.
- Apply invariant English culture for deterministic machine-readable formatting.
- Initialize current-result, reference-result, diagnostics, jobs, graph, and report state.

3.2 File processing phases

Phase	Action	Output
1. Selection	User selects or drops a file	Absolute source path
2. Native detection	Inspect header and identify container	NativeMediaInfo
3. Metadata preparation	Parse tracks, metadata, and sample tables/indexes	Container probe result
4. Canonical adaptation	Create common tracks and bounded samples	LsdMediaModel
5. Codec analysis	Read samples and parse frame/audio syntax	Diagnostics and statistics
6. Finalization	Build report, bitrate bins, quantizer counts	Current A result
7. Optional promotion	Set Ref B promotes completed A	Reference B snapshot
8. Comparison	Analyze a later A and render both results	A/B graphs and summaries

3.3 Background jobs

MP4/MOV and AVI preparation run in dedicated background threads. Codec analysis also runs in a background worker. The GUI timer polls progress and publishes completed objects. Reference B is created only from a completed displayed result; no background worker shares mutable histogram collections with the reference snapshot.

4. Canonical Media Model

4.1 LsdMediaModel

Field	Meaning
SourcePath	Input file path
ContainerType / ContainerDetail	Normalized family and specific route
FileLength	Source length in bytes
DurationSeconds	Best available duration
TimecodeScale	Nanoseconds per canonical timestamp tick
Tracks	Canonical video, audio, subtitle, and other tracks

4.2 LsdTrackModel and LsdSampleModel

Object	Key fields
LsdTrackModel	TrackId, TrackType, CodecId, CodecFamily, CodecPrivate, PayloadFormat, NalLengthSize, PCM parameters, Samples
LsdSampleModel	FileOffset, Length, DecodeTimestamp, PresentationTimestamp, Duration, key/invisible/discardable flags, PayloadFormat

4.3 Reference result snapshot

Reference B is a presentation-layer snapshot, not a second canonical media model. The snapshot contains the file identity, duration, bitrate bins, quantizer mode, quantizer counts, and a frozen Summary B text. Collections are copied at promotion time so later analyses cannot mutate B.

```
ReferenceResult = { File, Duration, BitrateBins[], QpMode, QpCounts{}, SummaryText }
```

5. Container Modules

5.1 Matroska and WebM

The Matroska reader parses EBML elements, Segment metadata, Tracks, Cluster timing, Block and SimpleBlock structures, and supported lacing modes. Canonical sample payload ranges are verified before traversal. MuxingApp and WritingApp are reported separately.

5.2 MP4, M4V, and MOV

The ISO Base Media reader supports non-fragmented files with a moov-based sample table. It derives sample offsets, sizes, decode times, composition offsets, and sync flags from stsd, stsz, stsc, stco/co64, stts, ctts, and stss. QuickTime sound sample entries and supported PCM forms remain unchanged from 2.0.

5.3 AVI 1.0 and OpenDML AVI 2.0

The AVI module reads RIFF headers, stream formats, LIST/INFO metadata, movi areas, legacy idx1 entries, AVIX segments, and standard ix### indexes. OpenDML indexes are preferred when valid; idx1 remains a controlled fallback. Payload offsets are deduplicated and bounded.

5.4 Comparison neutrality

Reference comparison never reads container structures. Once analysis completes, A and B are compared only through normalized output values. A Matroska-derived AVC result and an MP4-derived AVC result may therefore be displayed together, while the UI still exposes factual differences in their summaries.

6. Video Analysis Modules

Family	Native analysis
H.264 / AVC	avcC and Annex B support, SPS/PPS, I/P/B slices, native SliceQP, CABAC/CAVLC context, VUI metadata
H.265 / HEVC	hvcC, VPS/SPS/PPS context, NAL class, first-slice I/P/B, native SliceQP, VUI metadata
AV1	av1C validation, sequence/frame OBU traversal, frame classes, hidden/show-existing events, Base Q Index
MPEG-4 Part 2 / XviD	VOL/VOP traversal, I/P/B/non-coded VOPs, packed samples, VOP quantizer

6.1 Comparison rule for quantizers

A/B quantizer rendering is enabled only when the quantizer modes are compatible. AVC and HEVC use SliceQP, AV1 uses Base Q Index, and MPEG-4 Part 2 uses VOP quantizer. LSD does not present unlike quantities as if they were directly equivalent.

7. Audio Analysis Modules

Family	Validation and reporting
AAC	AudioSpecificConfig, core/output rate and channels, SBR, Parametric Stereo, AAC-LC/HE/HEv2 naming
MP3	Version/layer, bitrate and sample-rate index, padding, frame size, decoded samples, exact duration and bitrate
AC-3 / E-AC-3	Syncword, bitstream ID, sample rate, frame size, blocks, channels, LFE, decoded PCM samples
PCM	Block alignment, payload divisibility, decoded duration, expected and measured bitrate

Audio analysis remains single-result in the 3.0 graphs. Audio details for Reference B remain available through Summary B. No audio waveform or quality metric is synthesized.

8. Quantizer and Bitrate Analysis

8.1 Bitrate profile

Each canonical video sample contributes byte size to a fixed set of presentation-time bins. In comparison mode, Reference B is drawn in gold and Current A in blue. Both use a shared vertical maximum so differences remain visually meaningful. The result identity is role-based, not value-based: larger bitrate does not change a series color.

8.2 Quantizer sources

Codec	Native value	Histogram
AVC	SliceQPY	Frame-level SliceQPY
HEVC	SliceQPY	Frame-level SliceQPY
AV1	Base Q Index	Frame-level Base Q Index
MPEG-4 Part 2	VOP quantizer	Coded-VOP quantizer

8.3 Dual DRF/QP distribution

The right panel presents A and B statistics in aligned columns. For each quantizer value, two compact horizontal bars share one numeric row: blue A and gold B. Percentage columns are separately aligned and color-coded. The displayed range spans the first through last real value in either result plus one surrounding bucket on each side. This reduces empty visual space while preserving distribution context.

8.4 Completeness

- No estimated quantizer values are substituted.
- Rejected syntax units remain explicit.
- A comparison does not weaken per-result completeness rules.
- If quantizer modes differ, the graph reports comparison unavailable instead of forcing an overlay.

9. GUI, Jobs, Reporting, and Comparison

9.1 Views

View	Content
Summary A	Readable report for Current A
Summary B	Frozen readable report for Reference B; visible only while B exists
Streams	Structured current native-track data
JSON	Serializable current metadata and streams
Log	Current lifecycle and validation messages
Bitrate graph	Blue A and gold B with role legend
DRF/QP panel	A/B statistics, legend, bars, and percentages

9.2 Copy action

Copy follows the active report tab. When Summary A is active, Copy places A on the clipboard. When Summary B is active, Copy places the frozen reference report on the clipboard. Removing B safely returns selection to Summary A before removing the Summary B tab.

9.3 Presentation colors

Role	Color	Meaning
A Current	Blue	Most recently analyzed current result
B Reference	Gold	Stored comparison reference
Neutral UI	Gray / white	Grid, labels, inactive state, metadata

10. Reference Comparison State Model

The 3.0 workflow supports exactly one Current A and one Reference B. It is not a queue and does not attempt multi-file ranking.

```
EMPTY --Open--> A BLUE A BLUE --Set Ref B--> B GOLD B GOLD --Open--> B GOLD + A BLUE A BLUE + B GOLD --Remove  
Ref B--> A BLUE B GOLD --Remove Ref B--> EMPTY
```

10.1 State rules

- Every newly opened or dropped file becomes Current A and is blue.
- Set Ref B promotes the completed A result to B, changes its role color to gold, and waits for a new A.
- Opening another file while B exists preserves B and produces a new blue A.
- Remove Ref B removes only B. If A exists, A remains. If B is the only result, the window resets to initial state.
- B is never silently converted back into A after removal.

10.2 Reset behavior

When the last remaining B is removed, LSD clears Summary, Streams, JSON, Log, bitrate bins, quantizer counts, file labels, and copy availability; the status returns to Ready. The process and compiled C# types remain loaded, so this is a UI and result-state reset rather than an application restart.

11. Validation and Defensive Parsing

11.1 Boundary checks

- Canonical samples require non-negative offsets and positive lengths.
- Offset plus length must remain within the source file.
- Container box/chunk sizes are checked for overflow and parent-bound violations.
- NAL, OBU, MPEG Audio, Dolby, and PCM readers reject truncated payloads.
- Comparison snapshots are created only after completed analysis.

11.2 Error model

Optional metadata probes may fail without invalidating core analysis. Critical sample, index, and payload violations produce a controlled error or incomplete result. Diagnostic objects expose rejected counts and first failure. The comparison layer does not hide these diagnostics; Summary A and Summary B preserve each result independently.

11.3 Runtime isolation

- No external multimedia executable is launched.
- No temporary media files are created.
- The source file is opened read-only with shared read access.
- The single-file script can be hosted from memory by an EXE wrapper.

12. Supported Combinations

12.1 Container and video matrix

Video	Matroska/WebM	MP4/M4V/MOV	AVI
H.264 / AVC	Yes	Yes	Yes, Annex B
H.265 / HEVC	Yes	Yes	No
AV1	Yes	Yes	No
MPEG-4 Part 2 / XviD	Yes	Yes	Yes

12.2 Container and audio matrix

Audio	Matroska/WebM	MP4/M4V/MOV	AVI
AAC-LC / HE / HEv2	Yes	Yes	No
MP3	Yes	Yes	Yes
PCM integer / float	Yes	Supported entries	Yes
AC-3	Yes	Yes	Yes
E-AC-3	Yes	Yes	No

12.3 Comparison combinations

Bitrate comparison can display any two completed video results on a shared normalized x-axis. Quantizer comparison requires matching QpMode. Differences in codec family, dimensions, frame rate, duration, or frame count remain visible in Summary A and Summary B and must be interpreted by the user.

13. Known Limitations

- Fragmented MP4 using moof, traf, and trun is not supported.
- AVI H.264 targets Annex B payloads; unusual length-prefixed AVC in AVI may not be accepted.
- AVI AAC and AVI E-AC-3 mappings are not enabled.
- Container-only color metadata is not used as a fallback when the codec bitstream lacks color signaling.
- Certain HEVC PPS syntax involving scaling-list data remains intentionally incomplete.
- Reference comparison does not decode pixels and does not calculate PSNR, SSIM, or VMAF.
- Reference comparison supports one A and one B only.
- Bitrate differences across unlike resolution, frame rate, or duration are factual but not by themselves a quality verdict.

14. Verification and Regression Matrix

Scenario	Expected result
Open one supported file	Current A appears blue; Summary A and native graphs complete
Promote A to B	A role disappears; B remains gold; Summary B freezes
Open new file with B active	New A appears blue; B remains unchanged
Remove B while A exists	Gold B disappears; blue A and Summary A remain
Remove B when B is the only result	Window returns to startup state
Compatible QP modes	Dual DRF/QP statistics and bars appear
Different QP modes	Quantizer comparison unavailable; summaries remain accessible
Copy on Summary A / Summary B	Clipboard receives the active report

14.1 Required invariants

- Canonical sample counts equal validated indexes or sample tables.
- Packet/frame match is Yes for completed video analyses.
- Quantizer completeness remains PASS when one native value is obtained per relevant frame.
- Reference B values do not change after additional A analyses.
- Colors remain role-bound: A blue, B gold.
- Removing B cannot expose B again as A.

15. Maintenance Guidelines

15.1 Safe extension sequence

- Add or update container-native codec mapping.
- Normalize codec family and payload format.
- Populate canonical tracks and bounded samples.
- Reuse or isolate a payload reader and analyzer.
- Expose parsed/rejected counts and first failure.
- Add report fields only when authoritative data exist.
- Create deterministic regression assets for both new and existing paths.
- Keep comparison logic downstream of completed results.

15.2 Code that should remain stable

- Canonical model contracts and sample boundary validation.
- AVI/OpenDML chunk resolution and MP4 parent-bound checks.
- Matroska lacing accounting.
- Codec parser completeness rules.
- Background job cancellation and disposal.
- N/A reporting policy.
- A/B role semantics and snapshot immutability.

15.3 Future candidates

- Fragmented MP4 support.
- Container-color fallback through MP4 colr/nclx and Matroska Colour.
- Optional AVI AAC/E-AC-3 mapping.
- Additional video codecs such as VP9 or MPEG-2 Video.
- Optional textual compatibility summary for A versus B.

Appendix A. Key Types and Responsibilities

Type / state	Responsibility
NativeMediaInfo / NativeTrack	Container-native metadata and codec configuration
LsdMediaModel / LsdTrackModel / LsdSampleModel	Container-neutral tracks and bounded samples
NativeMediaProbe / NativeMp4Probe / NativeAviProbe	Container detection, metadata, and sample indexing
NativeMatroskaPacketScan	Shared canonical video analysis job; historical type name
CanonicalAudioAnalyzer	MP3, Dolby, and PCM validation/accounting
H264SliceHeaderParser / HevcHvccContextProbe	AVC/HEVC frame syntax and quantizer extraction
Av1CodecConfigurationProbe	AV1 configuration and sequence data
ReferenceResult	Frozen B file, bitrate bins, QP mode/counts, and summary
ReferenceOnlyView	Indicates that the promoted B is currently the only role
PowerShell coordinator	Jobs, state, reports, graphs, comparison, and GUI

Appendix B. Terminology

Term	Definition
Canonical sample	Container-neutral record describing a bounded payload range and timing
Codec private	Initialization data stored outside ordinary media samples
Payload format	Packaging of codec units inside a canonical sample
Current A	Most recently analyzed active result, rendered blue
Reference B	Stored comparison result, rendered gold
Promotion	Set Ref B transition that changes A into B
SliceQPY	Luma quantization parameter from AVC/HEVC slice syntax
Base Q Index	AV1 frame-level base quantizer index
Complete	All required units parsed without rejection for the relevant result

Appendix C. Release Identification

Property	Value
File name	LSD-3.0.ps1
Version	3.0 Final Release
SHA-256	8ec0c6f45f68426e186e88867ce60976922a3e68f41ce2f8f544a77d7fc9ac97
Runtime model	Single PowerShell file with embedded C#
Runtime dependencies	Windows PowerShell 5.1 and .NET Framework classes
External multimedia tools	None
Temporary media files	None

Release status. This document corresponds to the LSD 3.0 artifact identified above. Version 3.0 retains the stable canonical architecture and adds Current A / Reference B comparison, dual bitrate and quantizer views, reference summary retention, active-report copy behavior, and reset after removing the final result.